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Robot Control Parameters Tuning Using Bayesian optimization

Machine Learning for Mechanical Systems

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1 Introduction

We present the implementation of a Bayesian Optimization algorithm for the FRANKA Emika manipulator data driven control parameters tuning. BO is an efficient method in terms of performance and number of function evaluations for derivative-free optimization. Together with the robot model, the algorithm will compute internally a surrogate model of the problem, based on I/O and use this data-driven model to optimize the parameters for the particular application.

The aims of the project can be summarized in the following steps:

- **Optimization strategy comparison** :
  We compared a single general acquisition function optimization with a double step optimization with 2 different acquisition function Starting from the target trajectory we define a performance index which guides the control
- **Cost Function Definition** :
  Starting from the target trajectory we define a performance index which guides the control parameters tuning. In particular the aim is to optimize not only trajectory tracking performance, but also stability/safety and smoothness of the motion.
- **Sequential Joint Optimization**:
  Instead of optimizing all 21 PID parameters simultaneously, a sequential strategy is implemented. This approach proceeds by tuning the controller gains of each joint one at a time, beginning from the end-effector and moving backward along the kinematic chain. Such strategy reduces the dimensionality of the optimization problem and enhances computational efficiency, making the tuning process more manageable and scalable.
- **Mass Parameter Estimation**:
  The estimation of the equivalent link-mass parameters, which are involved in the feedback linearization and feedforward control components, is carried out through a Bayesian optimization procedure. This first phase is performed under open-loop control conditions and aims to identify the mass parameters by minimizing acceleration tracking errors derived from measured joint torques. Accurate mass estimation enhances the model-based compensation of gravitational and Coriolis effects, effectively decoupling the joint dynamics.

  Once the link-mass parameters are identified, a second optimization phase is performed to tune the joint-level PID gains, leveraging the improved model to refine trajectory tracking performance. In both stages, Bayesian optimization is applied to iteratively adjust the control parameters based on experimental evaluations.
- **Introduce Limit On Available Torque**:
  In order to ensure compliance with industrial hardware limitations, a constraint on the maximum torque applied by the actuators was introduced.

2 Optimization strategy

2.1. Optimization Strategy: Acquisition Function Analysis

In this chapter, we analyze the impact of changing the acquisition function on the efficiency of the Bayesian Optimization procedure, comparing two different strategies: a simple single-phase optimization and a two-phase approach that separates exploration and exploitation.

2.1.1. Single-phase Optimization using Expected Improvement

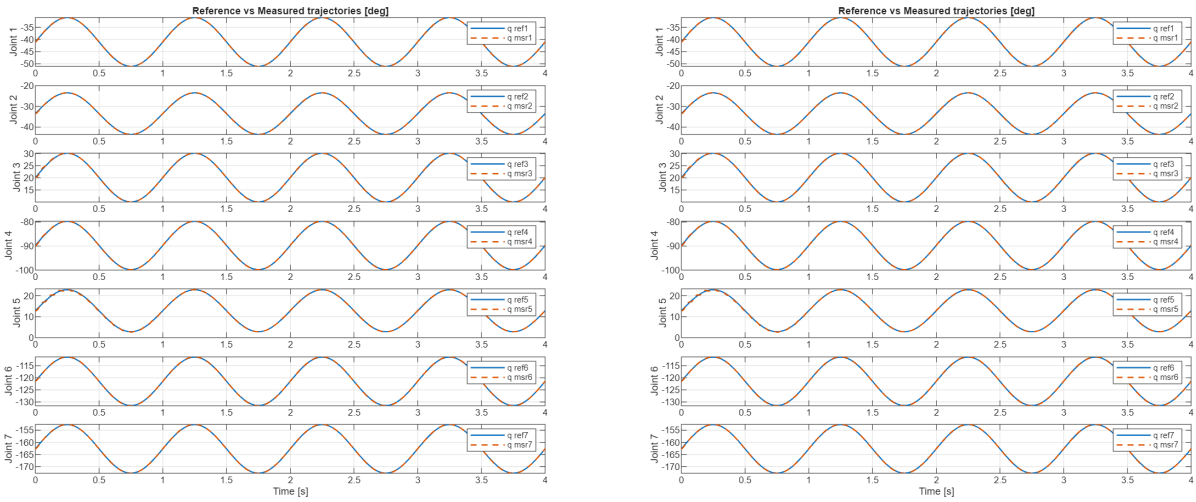
We first tested a one-phase strategy using the *Expected Improvement* (EI) acquisition function. This method is known to automatically balance exploration and exploitation based on predicted improvement, without the need for manual tuning of exploration weight.

The optimization was run using the hyperparameters shown in Table 1.

Table 1: Bayesian Optimization hyperparameters for Expected Improvement (EI) strategy

Parameter	**Value**
AcquisitionFunctionName	`expected-improvement`
Verbose	1
IsObjectiveDeterministic	`true`
UseParallel	`parallel_var`
MaxObjectiveEvaluations	230
MaxTime	`inf`
NumSeedPoints	15
GPActiveSetSize	300

The optimization converged smoothly and resulted in a solution with good position tracking accuracy. However, the velocity tracking performance,especially on joint 5, was significantly less accurate, highlighting a limitation of using EI alone in this context.



(a) Position tracking (b) Velocity tracking

Figure 1: Tracking performance of strategy 1

2.1.2. Two-phase Optimization: LCB + EI+

The second strategy splits the optimization into two distinct phases:

- **Phase 1: Exploration** — Uses the *Lower Confidence Bound* (LCB) acquisition function to prioritize broad search of the parameter space.
- **Phase 2: Exploitation** — Switches to *Expected Improvement Plus* (EI+) with an exploration ratio set to 0.1, focusing more on known high-performing regions.

The hyperparameters for both phases are reported in Tables 2 and 3 respectively.

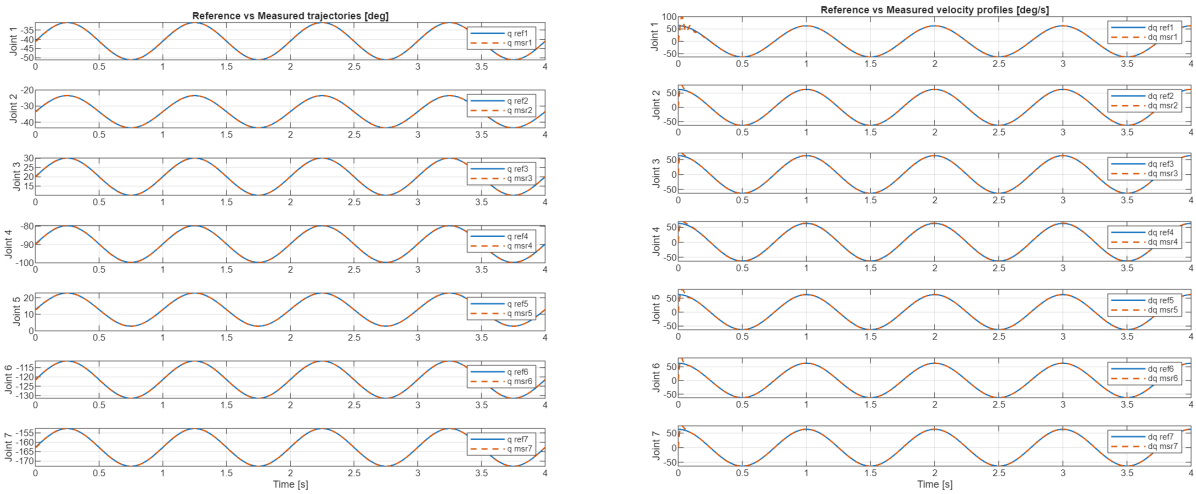
Table 2: Exploration phase hyperparameters (LCB)

Parameter	**Value**
AcquisitionFunctionName	`lower-confidence-bound`
Verbose	1
IsObjectiveDeterministic	`true`
UseParallel	`parallel_var`
MaxObjectiveEvaluations	150
MaxTime	`inf`
NumSeedPoints	15
GPActiveSetSize	300
PlotFcn	`[]`

Table 3: Exploitation phase hyperparameters (EI+)

Parameter	Value
AcquisitionFunctionName	`expected-improvement-plus`
ExplorationRatio	0.1
Verbose	1
IsObjectiveDeterministic	`true`
UseParallel	`parallel_var`
MaxObjectiveEvaluations	80
MaxTime	`inf`
InitialX	exporation set
InitialObjective	exploration objective set
GPActiveSetSize	300
PlotFcn	`[]`

The results of the tracking performance are much better especially for the velocity.



(a) Position tracking (b) Velocity tracking

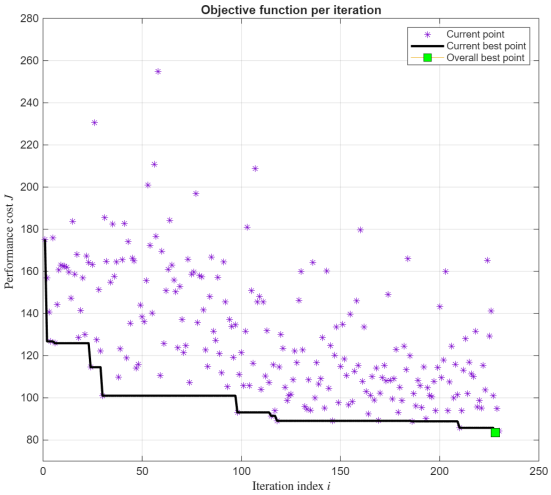
Figure 2: Tracking performance of strategy 1

2.1.3. Comparison and Observations

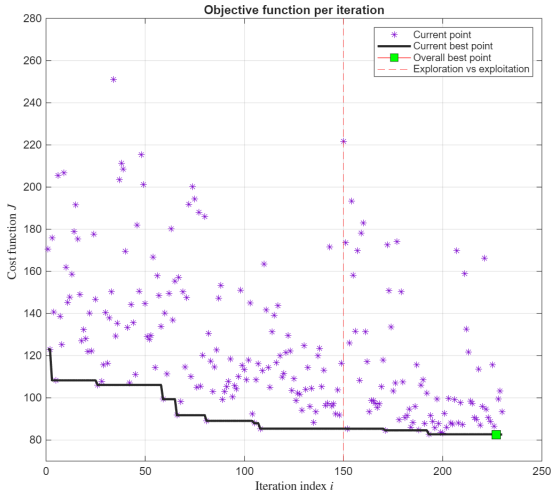
A first observation from both experiments is that in each case, the best solution was found near the end of the optimization process. While this might suggest delayed convergence, it is consistent with the principles of Bayesian Optimization, where the goal is not to find the minimum early, but rather to balance exploitation of promising regions with exploration of new ones, optimizing performance relative to evaluation budget.

More importantly, we find that the two-phase strategy clearly outperforms the single-phase one:

- The final solution obtained from the two-phase method yields better overall tracking performance.
- Notably, the second strategy had already outperformed the best solution of the single-phase EI method at iteration 193, achieving a cost value of $J = 82.7$.



(a) J evaluation – Strategy 1



(b) J evaluation – Strategy 2

Figure 3: Comparison of J evaluations over iterations for both strategies

Based on these results, all subsequent optimization procedures in this work will use the two-phase acquisition function strategy (LCB followed by EI+), as it provides a better tradeoff between exploration efficiency and exploitation accuracy.

3 Cost function definition

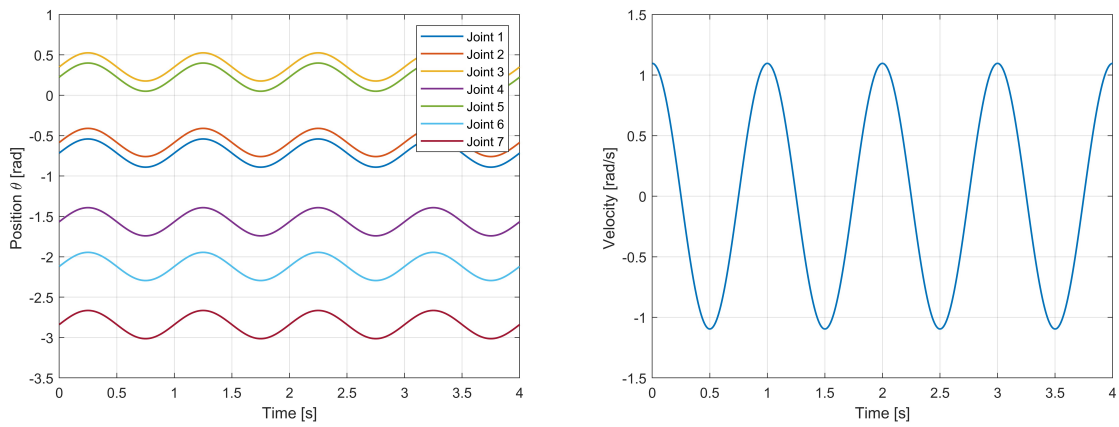
To guide the tuning of the control parameters, a cost index is formulated based on joint tracking errors in both position and velocity across a reference trajectory. Additional penalization terms are introduced for parameter combinations that produce excessive deviations or lead to unstable or unsafe conditions, causing the simulation to terminate prematurely.

Two distinct formulations are employed:

- The first one assess the performance of each motor.
- The second on evaluates the performance of the end effector only

The present chapter focuses on the former, aiming at the optimization of PID gains at the joint level. The robot's inertial parameters are assumed to be known for this process.

The reference joint profiles are synthesized through periodic excitation signals for each degree of freedom. These signals are illustrated below, which includes both position and shared velocity profiles:



(a) (b)

Figure 4: Target cinematic reference a) trajectory reference profile for each joint, b) velocity reference profile shared between each joint

For each iteration of the optimization routine, the system dynamics are simulated using the current PID parameters, and the resulting trajectory is compared against the reference to compute a cost value.

3.1. combined joint's error cost function

The cost function $\mathcal{J}$ used throughout this study is defined as a weighted sum of normalized performance terms, quantifies the tracking accuracy, smoothness, and actuation effort across all seven joints. It is computed as:

$$J = \sum_{j=1}^{Nmetrics} \sum_{i=1}^{7} w_j e_{ij} + P \tag{1}$$

Here, ε_{ij} represents the normalized tracking and effort-related metric j for joint i, while w_j is a weight vector assigning importance to each metric.
Each metric in ε_i is normalized and the weights assigned to each metric are the following:

Symbol	Description	Formula	Weight
e^i_{max}	Max. normalized joint position error	$\max_t \left\lvert \frac{q_{\text{ref},i}(t) - q_i(t)}{q_{r,\text{amp}}} \right\rvert$	50
$\dot{e}^i_{\text{max}}$	Max. normalized joint velocity error	$\max_t \left\lvert \frac{\dot{q}_{\text{ref},i}(t) - \dot{q}_i(t)}{\dot{q}_{r,\text{amp}}} \right\rvert$	5
e^i_{avg}	Avg. normalized joint position error	$\frac{1}{T}\int_0^T \left\lvert \frac{q_{\text{ref},i}(t) - q_i(t)}{q_{r,\text{amp}}} \right\rvert dt$	30
$\dot{e}^i_{\text{avg}}$	Avg. normalized joint velocity error	$\frac{1}{T}\int_0^T \left\lvert \frac{\dot{q}_{\text{ref},i}(t) - \dot{q}_i(t)}{\dot{q}_{r,\text{amp}}} \right\rvert dt$	10
j^i_{avg}	Avg. normalized joint jerk error	$\frac{1}{T}\int_0^T \left\lvert \frac{\dddot{q}_{\text{ref},i}(t) - \dddot{q}_i(t)}{\dddot{q}_{r,\text{amp}}} \right\rvert dt$	2.5
τ^i_{max}	Max. normalized applied torque	$\max_t \left\lvert \frac{\tau_{\text{PID},i}(t) + \tau_{\text{comp},i}(t)}{\tau_{\text{max},i}} \right\rvert$	2.5

Table 4: Definition of normalized joint metrics and their associated cost weights.

The normalization constants $q_{r,\text{amp}}$, $\dot{q}_{r,\text{amp}}$, and $\dddot{q}_{r,\text{amp}}$ correspond to the amplitude of reference profiles in position, velocity, and jerk. Torque normalization uses the actuator-specific bound $\tau_{\text{max},i}$ that are reported in the saturation chapter.

To account for critical deviations, the function includes a penalty term P which assumes the value P=300 when:

- The condition number of the Jacobian matrix exceeds a limit:
  $cond(Jac) > 10^6$
  Then a singularity is considered to have occurred. In particular it indicates the manipulator is near a kinematic singularity, where small changes in joint space may cause large or undefined changes in the end-effector's position/velocity. This can lead to instability or loss of control.
- if joint position errors exceed 15° to exclude poorly performing solutions.

This cost formulation supports multi-objective tuning, considering accuracy, smoothness, and actuation limits. It is iteratively evaluated by the Bayesian Optimization process to refine PID parameters.

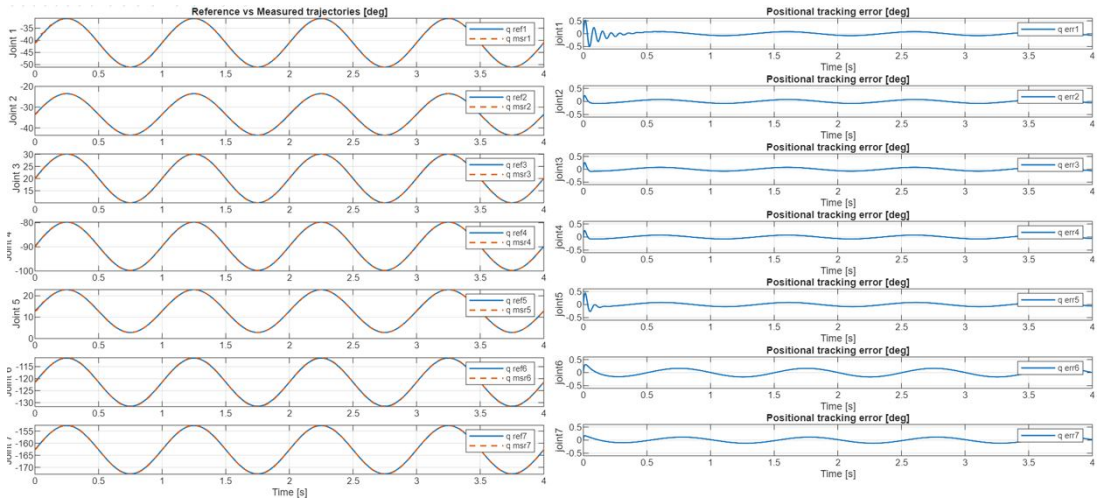


Figure 5: CF 1 position error

3.2. End effector error cost function

To broaden the scope of the controller optimization and better align with real-world operational demands, a second cost function was introduced as a complementary approach. This second formulation focus on the end effector's trajectory accuracy. This choice is driven by practical use cases where the quality of task execution is primarily judged by the precision of the tool center point rather than the detailed dynamics of individual joints.

In this alternative setup, the performance index penalizes deviations with respect to the reference in position, velocity, and jerk of the end-effector in cartesian coordinates. By doing so, it directly captures the accuracy and smoothness of the robot's motion in the workspace. This task space oriented formulation is particularly relevant when tuning all joint control gains jointly, as it allows the optimization process to holistically coordinate joint behaviors to serve a global objective: minimizing tracking error where it matters most, at the point of interaction with the environment.

The second cost function is defined as:

$$\mathcal{J} = \sum_{j=1}^{Nmetrics} \sum_{i=1}^{3} w_j \varepsilon_{ij} + \mathcal{P} \tag{2}$$

Here, ε_{ij} denotes the normalized metric j evaluated along Cartesian direction i (i.e., x, y, z), and w_j is the associated weight. Each metric captures a different aspect of task-space performance, including position and velocity tracking, jerk smoothness, and control effort.

The full list of considered metrics is reported below:

Symbol	Description	Formula	Weight
$e^i_{\max}$	Max. normalized cartesian position error	$\max_t \left\lvert \frac{s_{r,i}(t) - s_{msr,i}(t)}{s_{r,i,\text{amp}}} \right\rvert$	50
$\dot{e}^i_{\max}$	Max. normalized cartesian velocity error	$\max_t \left\lvert \frac{\dot{s}r,i(t) - \dot{s}msr,i(t)}{\dot{s}r,i,\text{amp}} \right\rvert$	5
$e^i\text{avg}$	Avg. normalized cartesian position error	$\frac{1}{T}\int_0^T \left\lvert \frac{s_{r,i}(t) - s_{msr,i}(t)}{s_{r,i,\text{amp}}} \right\rvert dt$	30
$\dot{e}^i_{\text{avg}}$	Avg. normalized cartesian velocity error	$\frac{1}{T}\int_0^T \left\lvert \frac{\dot{s}r,i(t) - \dot{s}msr,i(t)}{\dot{s}r,i,\text{amp}} \right\rvert dt$	10
$j^i\text{avg}$	Avg. normalized cartesian jerk error	$\frac{1}{T}\int_0^T \left\lvert \frac{\dddot{s}r,i(t) - \dddot{s}msr,i(t)}{\dddot{s}r,i,\text{amp}} \right\rvert dt$	2.5
$\tau^i\max$	Max. normalized applied torque	$\max_t \left\lvert \frac{\tau_{\text{PID},i}(t) + \tau_{\text{comp},i}(t)}{\tau_{\max,i}} \right\rvert$	2.5

Table 5: Definition of normalized end-effector metrics and their associated cost weights.

The normalization constants $s_{r,i,\text{amp}}$, $\dot{s}r,i,\text{amp}$, and $\dddot{s}r,i,\text{amp}$ refer to the expected amplitude of the reference signals in position, velocity, and jerk along each Cartesian axis i. Torque normalization uses actuator-specific limits $\tau_{\max,i}$.

To penalize unsafe behavior or large tracking deviations, a penalty term is added to the cost function which is the same as the first case for what concerns the Jacobian , while the threshold for poorly performance at a end effector position errors above 0.03 m.

The optimization hyper parameters and the evaluation metrics are the same of the first cost function.

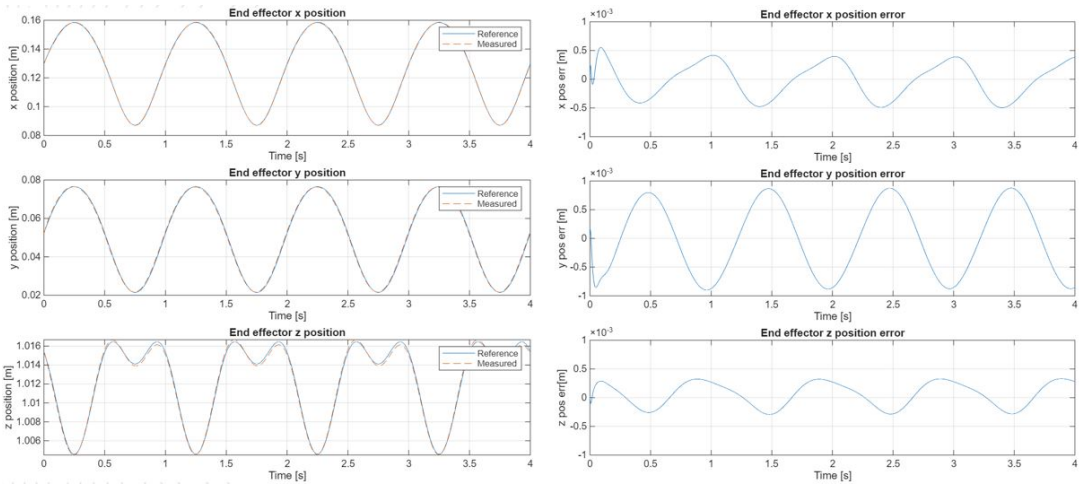


Figure 6: CF 2 position error

3.3. Evaluation of Cost Function Performance

In order to compare the effectiveness of two different cost function formulations, we defined a set of performance metrics that quantify the tracking behavior of the robot. These metrics are computed for each joint of the manipulator based on the measured joint trajectories and their respective references. The metrics are focused on steady-state performance and transient response.

Let $q_{\text{ref}}(t) \in \mathbb{R}^{n_{\text{dof}} \times T}$ be the matrix of reference joint positions, and $q_{\text{meas}}(t) \in \mathbb{R}^{n_{\text{dof}} \times T}$ the measured positions. The following metrics are computed for each degree of freedom:

3.3.1. Steady-State RMS Error

The Root Mean Square error over the final N cycles of the trajectory is calculated as:

$$\text{RMS}_i = \sqrt{\frac{1}{N} \sum_{k=1}^{N} (q_{\text{ref},i}(k) - q_{\text{meas},i}(k))^2}$$

This metric quantifies the average quadratic deviation in the steady-state region.

3.3.2. Overshoot Percentage

The overshoot is computed based on the signal amplitude during the steady-state window:

$$\text{Overshoot}_i = \max\left(\frac{A_{\text{meas},i} - A_{\text{ref},i}}{A_{\text{ref},i}} \cdot 100,\ 0\right)$$

where $A = \max(q) - \min(q)$ is the signal amplitude. This metric measures how much the system exceeds the amplitude of the reference trajectory.

3.3.3. Settling Time

Settling time is defined as the time after which the absolute tracking error remains below a given threshold $\varepsilon = 0.1°$ which corresponds roughly to $\varepsilon = 0.5mm$ positional tolerance, depending on

the joint's length., for at least two consecutive periods of the input signal:

$$|q_{\text{ref},i}(t) - q_{\text{meas},i}(t)| < \varepsilon, \quad \text{for 1 periods}$$

If this condition is never met during the simulation, the joint is considered "not settled."

3.3.4. results

Table 6: Performance comparison between CF1 (joint-based) and CF2 (end-effector-based)

KPI	CF1 (joints avg)	CF2 (coord avg)
Steady-state RMS error	0.001 rad	0.00037 m
Overshoot	0.88 %	0.85 %
Settling time	0.26 s	0.12 s

CF1 PID gains

Table 7: PID gains from CF1 (joint-based cost function)

Gain	Joint 1	Joint 2	Joint 3	Joint 4	Joint 5	Joint 6	Joint 7
K_p	8576.5	7478.9	8067.9	8195.9	7228.5	1897.1	1942.0
K_i	237.67	1048.1	421.08	998.55	58.26	740.31	946.80
K_d	18.272	157.84	127.99	134.73	49.465	141.49	247.48

CF2 PID gains

Table 8: PID gains from CF2 (end-effector cost function)

Gain	Joint 1	Joint 2	Joint 3	Joint 4	Joint 5	Joint 6	Joint 7
K_p	2314.4	1005.0	7781.2	1183.8	2120.3	9841.2	808.97
K_i	309.83	164.92	758.38	955.49	737.80	643.99	918.99
K_d	77.205	268.02	317.72	299.05	61.265	228.65	317.46

we can see that the results are comparable, however the point of these analysis is having the option to prioritise two different POVs.

The joint-based cost function provides a more direct evaluation of internal control performance, whereas the end-effector-based formulation is more task-oriented and aligned with operational requirements.

While both strategies led to stable controllers, the TCP-based cost function produced finer precision in Cartesian space, with lower steady-state errors and slightly faster settling behavior. However, one of the Cartesian coordinates did not settle within the simulation time, suggesting sensitivity to tuning.

As a result, future development may consider hybrid cost functions or adaptive strategies that combine both perspectives, depending on task requirements.

4 | Torque Saturation And Physically Feasible Trajectory

In order to ensure physical feasibility and compliance with industrial hardware limitations, a constraint on the maximum torque applied by the actuators was introduced. This bound was selected based on typical industrial specifications for joint torque limits and serves to prevent unsafe or unrealistic motor commands during the tuning process.

The system is now constrained to apply, at most, the specified torque values, maintaining the maximum value in cases where the required torque exceeds the limit.

It can be observed that the torques applied by the motors exceed the nominal limit values (tab. 9). To increase realism, actuator saturation was introduced by limiting the maximum torque for each joint, based on real robot's physical specifications.

	Joint 1	**Joint 2**	**Joint 3**	**Joint 4**	**Joint 5**	**Joint 6**	**Joint 7**
Max torque	87 Nm	87 Nm	87 Nm	87 Nm	12 Nm	12 Nm	12 Nm

Table 9: Maximum torque values for each joint

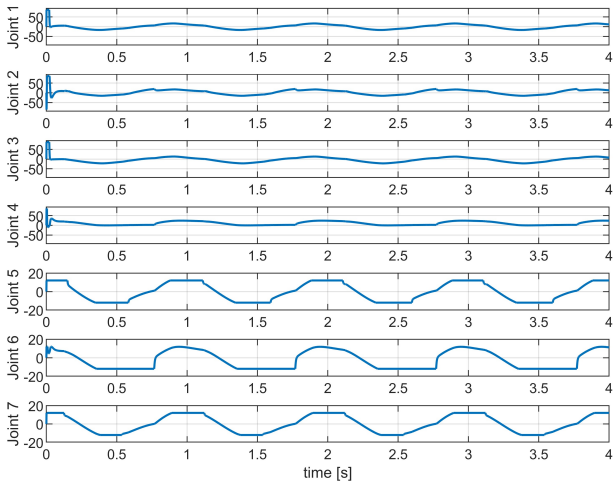


Figure 7: Joint torques profiles

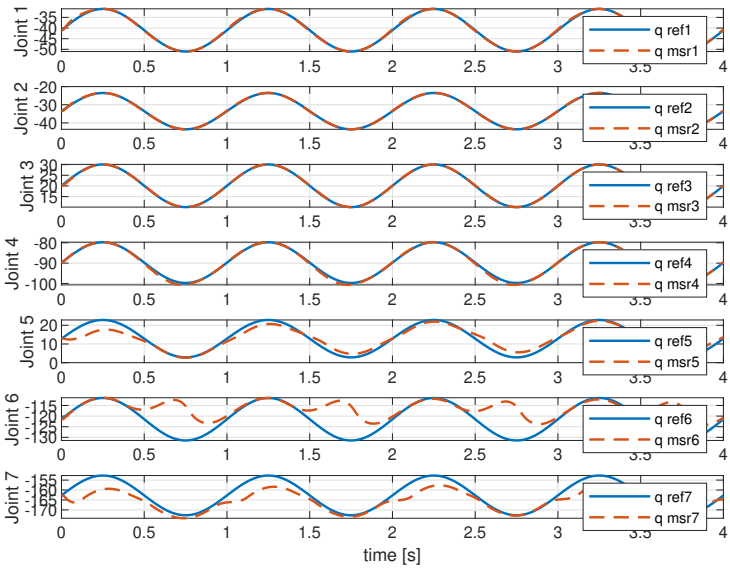


Figure 8: trajectory profile for each joint compared against reference

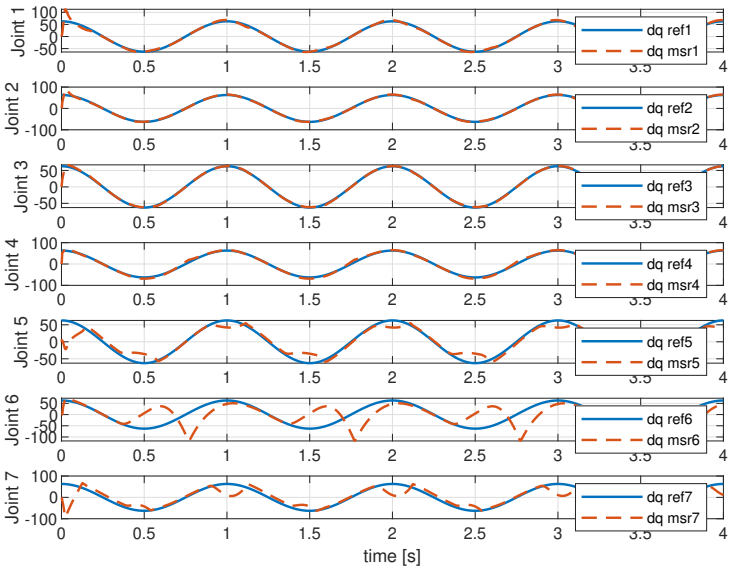


Figure 9: Velocity profile for each joint compared against reference

It can be observed that while the first four joints show results similar to the baseline case, the last three joints (which are subject to a torque limit of 12 Nm) are unable to accurately follow the target references. This occurs because the torques required to achieve those references exceed the imposed limit, resulting in errors in terms of motion amplitude and delays in the response.

A remark should be made about the velocity profiles: for the first four joints, at the beginning of the motion, an initial overshooting in velocity occurs. However, this overshoot is quickly corrected, and the velocity then closely follows the reference. Also the last three joints present a similar error in initial motion, but however they aren't able to follow the reference due to limitations on the torque to be applied.

5 | Cascade Bayesian Optimization

5.1. Methodological Approach

The script adopts a staged optimization approach aimed at efficiently tuning the PID parameters. The key strategy lies in breaking down the high-dimensional optimization task by addressing one joint at a time. This decomposition reduces the dimensionality of the search space, enhancing both tractability and convergence speed.

The tuning process progresses sequentially, beginning with the joint closest to the end-effector. At each stage, optimization targets only one joint, while the previously tuned joints retain their optimized parameters. The remaining joints, yet to be tuned, are temporarily assigned nominal values (typically zero), thereby minimizing their impact on the cost function. This sequential isolation allows the Bayesian Optimization (BO) algorithm to operate within a lower-dimensional parameter space during each phase, gradually constructing a complete controller.

Although this method assumes independent tuning of joints, such an assumption doesn't fully align with the physical behavior of serial manipulators. In practice, joint dynamics are interdependent due to shared inertial loads, Coriolis interactions, and gravitational effects. Nevertheless, under certain conditions, this decoupled approach can serve as a valid simplification—particularly when unoptimized joints are assigned small, non-disruptive gains and when the reference trajectories evolve smoothly. In this context, the residual inter-joint couplings are largely managed by the local PID controllers. While this introduces some level of suboptimality, it significantly simplifies the optimization, making the framework scalable and practically effective.

The hyperparameters chosen for each one of the seven consecutive BO are defined as follows to have comparable results with the other results:

- N° of initial seed points: 10
- N° of iterations in exploration phase: 50
- N° of iterations in exploitation phase: 15

The cost function adopted for this optimization process is the one previously defined as joint-space cost function.

5.2. Results

Here are shown the results for the cascade BO.

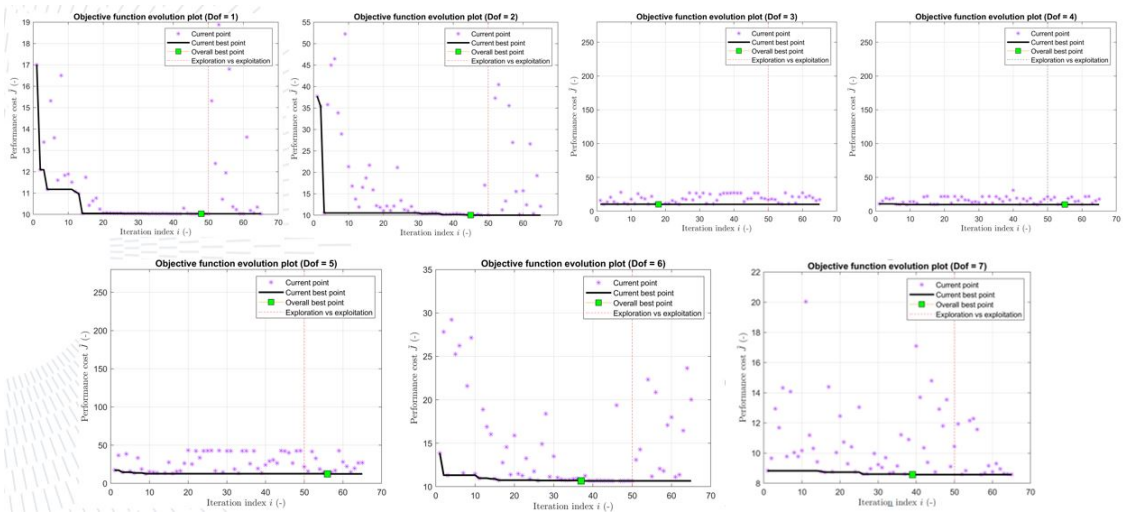


Figure 10: Cascade J per iteration per joint

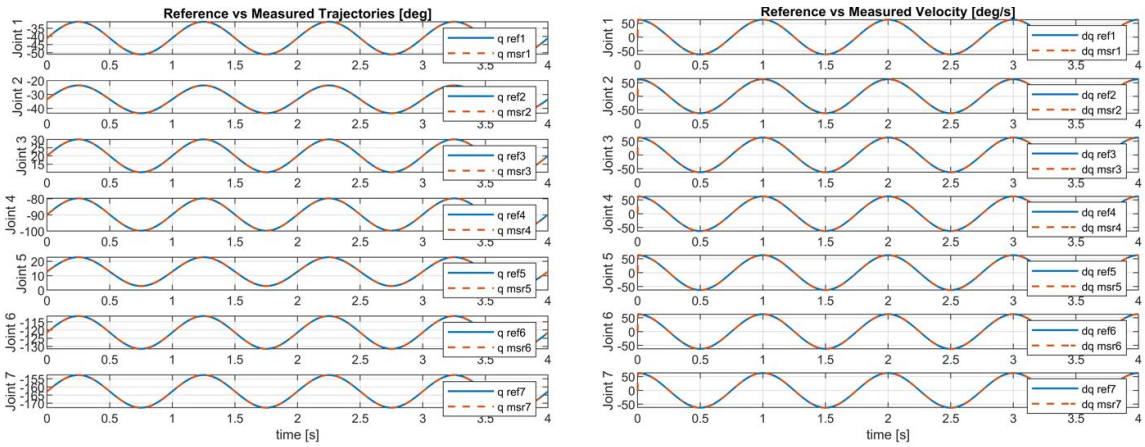


Figure 11: Cascade positional tracking and velocity tracking performance

The Pid gains are the following

	Joint 1	**Joint 2**	**Joint 3**	**Joint 4**	**Joint 5**	**Joint 6**	**Joint 7**
K_p	728.5	10985	752.4	995.2	659.1	882.7	846.9
K_i	297.2	351.0	70.3	66.8	3.25	32.1	59.0
K_d	553.6	429.3	546.7	550.1	502.4	497.5	498.9

Table 10: PID gains for cascade optimization preserving robot performance

6 Mass Estimation

6.1. Mass Identification and Gain Optimization Framework

In real-world robotic systems, the exact values of link masses are often unknown or only approximately available. This imprecision particularly affects model-based control strategies such as feedback linearization, which rely on accurate dynamic models to effectively counteract gravitational and Coriolis forces. Misestimating these parameters can compromise feedforward compensation, resulting in poor trajectory tracking and increased control effort.

To mitigate this issue, we adopt a two-phase optimization strategy. The first phase focuses on estimating the mass distribution of the robot links by exploiting the inverse dynamics model. We assume a control input structured as:

$$\boldsymbol{\tau} = \mathbf{B}(\mathbf{q}, \mathbf{m})\ddot{\mathbf{q}}_d + \mathbf{C}(\mathbf{q}, \dot{\mathbf{q}}, \mathbf{m})\dot{\mathbf{q}} + \mathbf{g}(\mathbf{q}, \mathbf{m}) \tag{3}$$

When the estimated mass parameters $\mathbf{m}$ are accurate, the dynamics of the real system match those of the model, and the following equality holds:

$$\mathbf{B}(\mathbf{q})\ddot{\mathbf{q}} + \mathbf{C}(\mathbf{q}, \dot{\mathbf{q}})\dot{\mathbf{q}} + \mathbf{g}(\mathbf{q}) = \mathbf{B}(\mathbf{q}, \mathbf{m})\ddot{\mathbf{q}}_d + \mathbf{C}(\mathbf{q}, \dot{\mathbf{q}}, \mathbf{m})\dot{\mathbf{q}} + \mathbf{g}(\mathbf{q}, \mathbf{m}) \quad \Rightarrow \quad \ddot{\mathbf{q}} = \ddot{\mathbf{q}}_d \tag{4}$$

The estimation is framed as an optimization problem, where a cost function is minimized that penalizes the difference between measured joint accelerations $\ddot{\mathbf{q}}$ and their desired counterparts $\ddot{\mathbf{q}}_d$. Bayesian Optimization (BO) is employed for this purpose, enabling efficient exploration of the parameter space with a reduced number of evaluations.

Accurate mass estimation significantly enhances the performance of feedforward terms in the control law, which in turn improves tracking quality and reduces the complexity of subsequent gain tuning.

The mass identification process uses the following BO hyperparameters:

- Number of initial seed evaluations: 15
- Number of iterations for exploration: 150
- Number of iterations for exploitation: 80

After completing mass estimation, a second optimization is performed to tune the PID control gains using a single-phase Bayesian Optimization approach. The corresponding hyperparameters are:

- Number of initial seed evaluations: 15
- Number of iterations for exploration: 150
- Number of iterations for exploitation: 80

the results are the following.

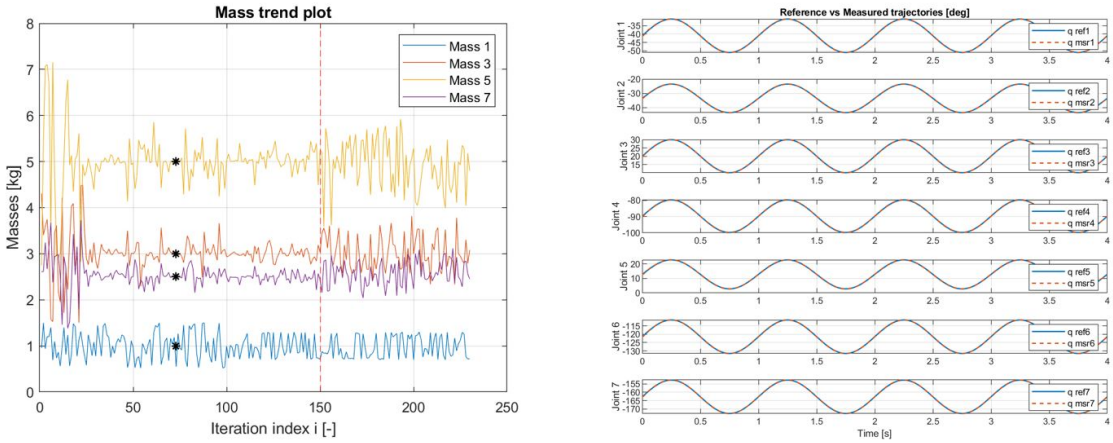


Figure 12: mass candidates evolution , trajectory tracking performance

The estimated masses are the following:

Joint	Joint 1	Joint 2	Joint 3	Joint 4	Joint 5	Joint 6	Joint 7
Real mass [kg]	1.0000	0	3.0000	0	5.0000	0	2.5000
Estimated mass [kg]	1.0027	/	2.9981	/	5.0042	/	2.4986

Table 11: Real and estimated mass values for each joint

As shown in table, this optimization method proves effective not only for tuning the PID gains, but also for estimating the inertias of each joint. Morover the algorithm is again able to faithfully reproduce the assigned reference trajectories even in the presence of small variations in the mass values compared to the real ones. This is due to the effectiveness of the controller, which is either insensitive to such small variations or capable of compensating for these minor mass estimation errors.